

Soptik Sir
Biology

"notes by Varbhav Bivush"

Nervous System

(CNS)
Central Nervous System
 (Brain and spinal chord)

(Integrating, processing and coordinating sensory data and motor commands)

Peripheral Nervous System
 (all the neural tissues outside the CNS)

Autonomic

(Communicates with internal organs and glands)

Sympathetic div.
 (arousing)

Parasympathetic div.
 (calming)

Somatic

(Communicate with sense organs and voluntary muscles)

Sensory (afferent)
 (Sensory input)

Motor (efferent)
 (Motor output)

Function → main (3)

Sensory Integrative Motor

CNS

③ Info processing include integration and distribution of info in CNS

↓

④ The motor division of PNS carries motor command from CNS to Per-tissues and Syst.

Includes

Somatic (SNS)
 (Controls skeletal muscle contraction)

Autonomic (ANS)
 provides automatic regulation of smooth muscle, cardiac muscle, glands, adipose)

② The sensory division of PNS brings info to CNS from receptors in peripheral tissues & organs

Somatic
 (pos., touch, pressure, pain, temp)

Special
 (smell, taste, vision, balance, hearing)

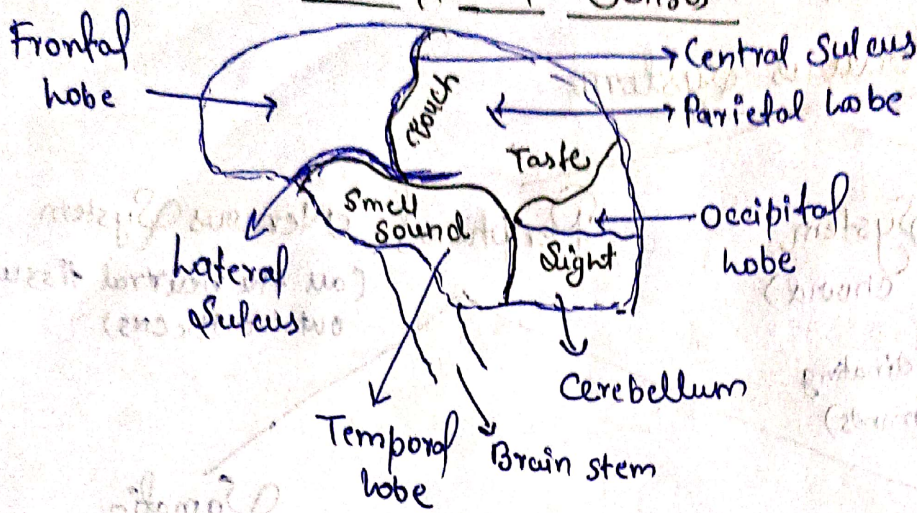
Visceral
 (Monitor internal organ)

Sensory receptors

① **Receptors** — detect change

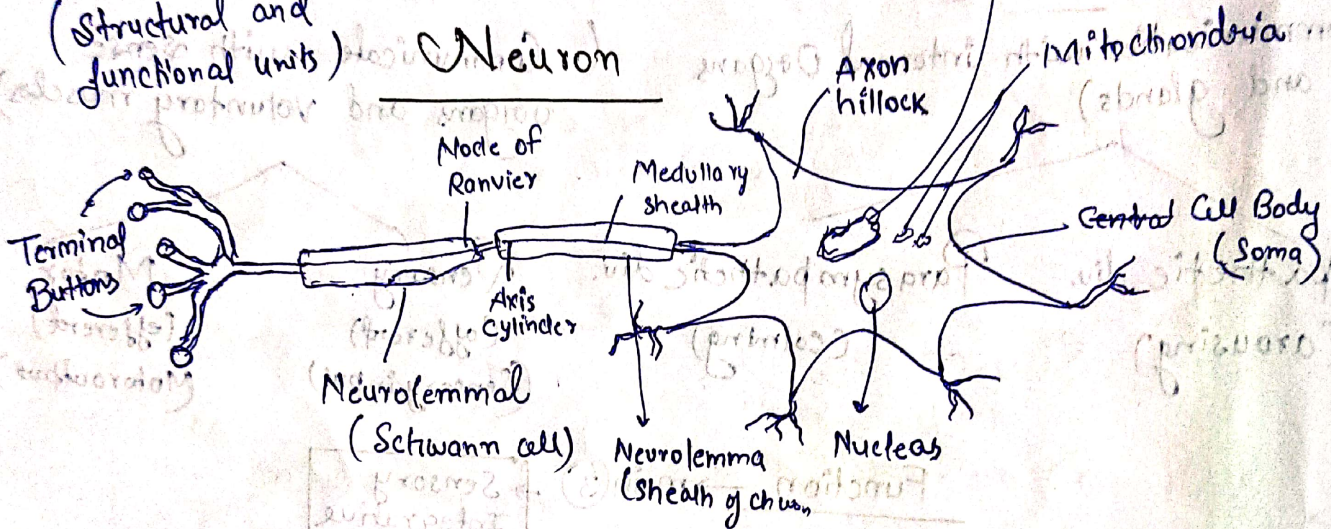
⑤ **Effectors** — Target organs whose activities change in response to neural commands.

The special Senses



(Structural and functional units)

Neuron



Nervous System

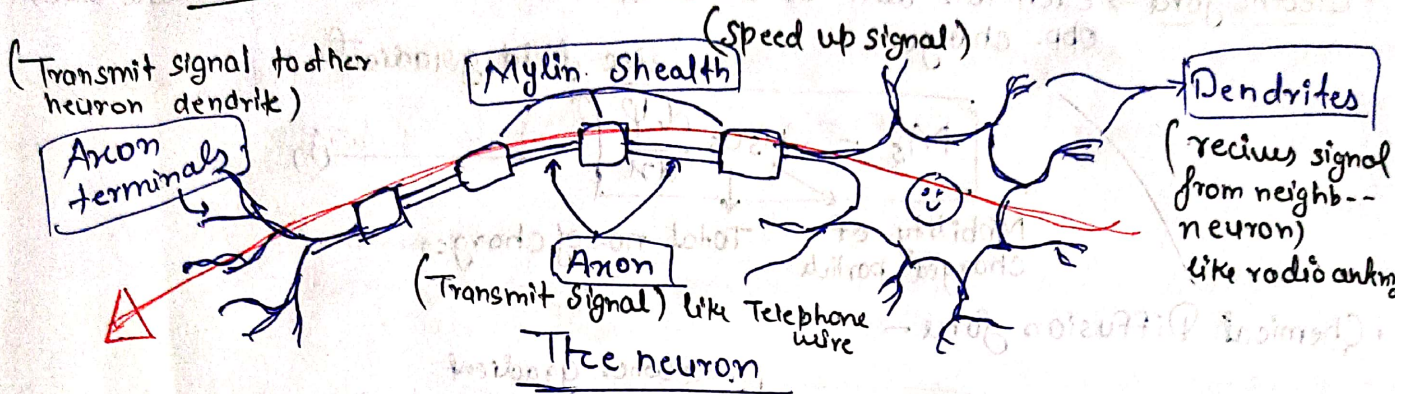
Nervous Tissue

Neuron

Neuroglia

Communicate through Synapse (Impulse via chemical and electric signals)

Nerve Impulse and Muscle Contraction



Neuron - cell that transfer information.

Potential Difference / Resting Potential

$\text{Na}^+ \text{Na}^+ \text{Na}^+$

$\text{V}_{\text{gC}} \rightarrow$ Voltage gated channel

$\text{Na-K Pump} \rightarrow$ Always open.

Concept

Outside 3Na^+ , Inside 2K^+
 Na^+ +ve charge, K^+ -ve charge
 (-70mv) (Polarised state)

When a nerve cell is at 'rest', membrane potential $\rightarrow$ Resting potential.

2 forces $\rightarrow$ Diffusion force
 Electric force

When impulse come,

$\text{Na-V}_{\text{gC}} \rightarrow$ open, and 2Na^+ come in

So, Out $\rightarrow +ve$ in $\rightarrow -ve$
 (Depolarisation)

At this time, potential $\rightarrow$ **Action potential**
 (-50mv)

To make normal,

K-V_{gC} open, 2K^+ go outside.

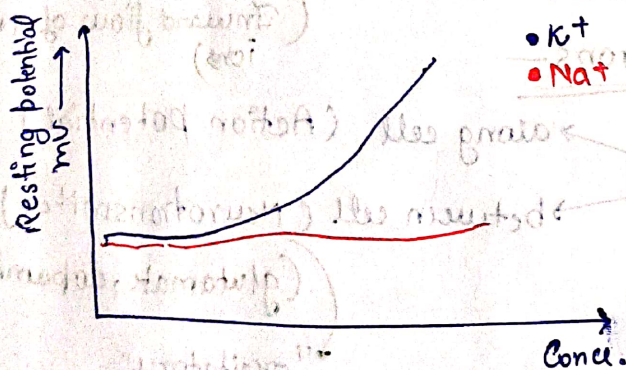
(Repolarisation)

To make full normal,

Na-K pump works,

2K^+ come in, 2Na^+ go out.
 (Polarised state)

Repeat



Action Potential info - (occurs in axon)

AP speed $\rightarrow$ 100 m/s

Size Varies neuron to neuron

Myelinated axon $\rightarrow$ AP faster

Do not decay with time and distance (constant)

Resting potential expression -

- Electric force → each ion will be attracted to the side of membrane with opp. charge.

$$M_s' = -u_s c_s \frac{d\psi}{dx} \quad (i)$$

Mobility of charged particle

Total no. of charges

- Chemical Diffusion force -

$$M_s' = -D_s \frac{dc_s}{dx} \quad (ii)$$

Diffusion constant

- Each ions are acted by two forces combinedly called Electrochemical forces

By (i) and (ii).

$$M_s' = -u_s c_s \frac{d\psi}{dx} - D_s \frac{dc_s}{dx}$$

At equilibrium,

$$M_s = 0$$

Simplify → we get membrane potential (Nernst equation)

$$E_x = \frac{RT}{nF} \ln \frac{X_o}{X_i}$$

Signal Propagation Between Neurons -

(Inward flow of Na^+ ions)

Two process of sending signals

- along cell (Action potential)
- between cell (Neurotransmitter)

(glutamate, dopamine)

The Sarcomere

fundamental unit of
Skeletal Muscle

"excitatory" firing a target neuron
(glutamate)

"inhibitory" makes a target neuron less likely to fire (GABA)

Structure of Muscles (like Towels में Pipes के
(Skeletal Muscles) में Pipes के - - - -)

(like Truck में Pipes के
अन्तर pipes के ----)

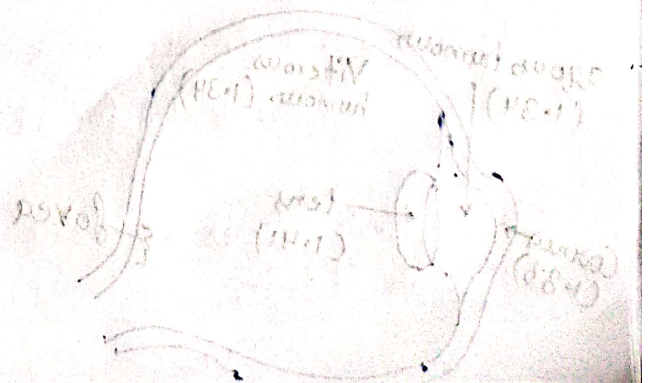
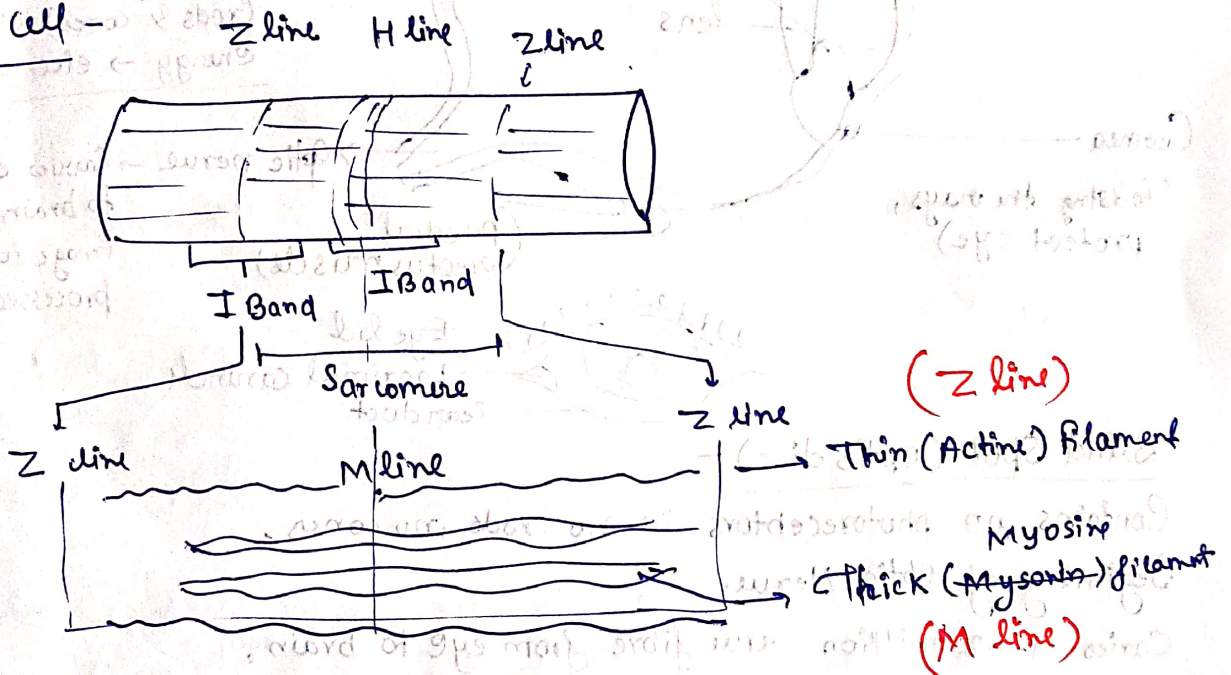
(Sarcomere)

Muscle $\rightarrow$ Muscle Bundle $\rightarrow$ Muscle fiber $\rightarrow$ Myofibrils $\rightarrow$ Myofilaments
 fasci (fascicles) (muscle cell) (zoom) (long proteins)

(Surrounded by connective tissue called Perimysium)

Skeletal muscle are usually attached to bone by Tendons composed of connective tissue $\rightarrow$ also make muscle called (epimysium).

Muscle cell



Class - 3

Receptors

Mechanoreceptor

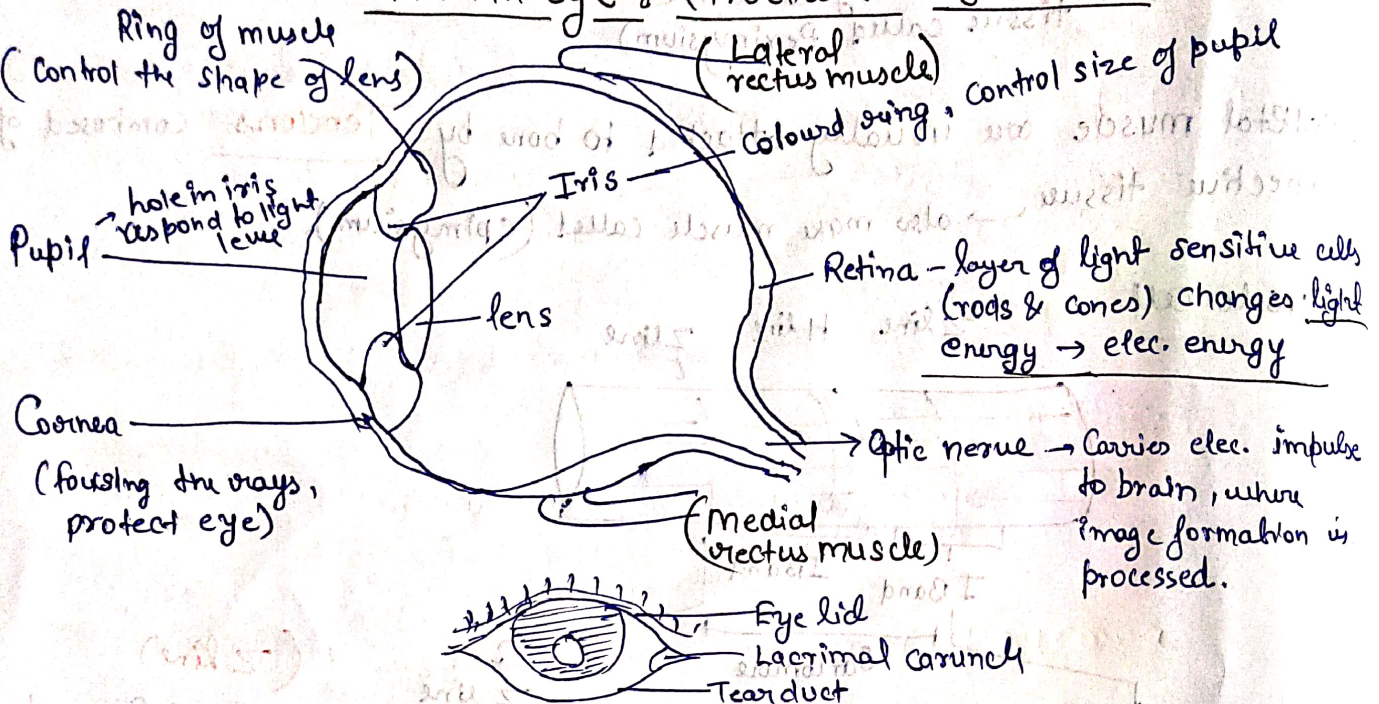
Thermo
(Cold/warm)

Nociceptors
(pain)

Elec. Mag.
(light)

Chemo.
taste, smell
O₂ level in blood
CO₂ conc.

Human eye : Structure & function



Blind Spot (Optic disc) -

Contains no photoreceptors i.e. no rods, no cones.

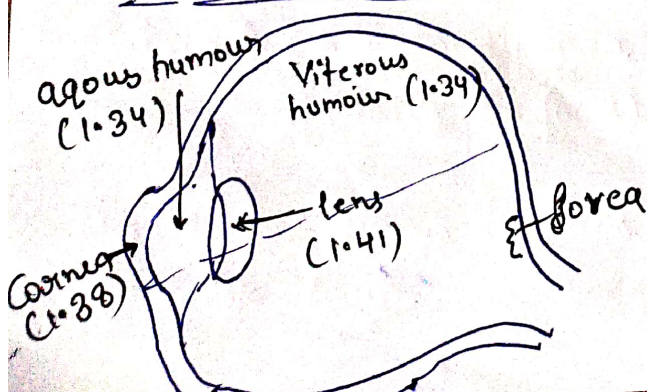
Beginning of Optic Nerve.

Carries 1-1.2 million nerve fibre from eye to brain.

The Eye (as a camera) -

- ① Lens System
- ② Aperture System (pupil)
- ③ A film (Retina)

Lens system Contains 4 refractive index



$$RI = \frac{\sin \theta_1}{\sin \theta_2}$$

The electromagnetic Spectrum

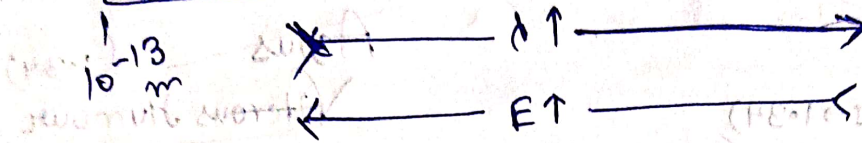
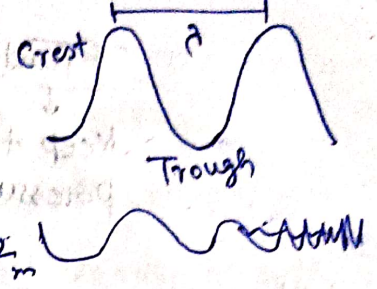
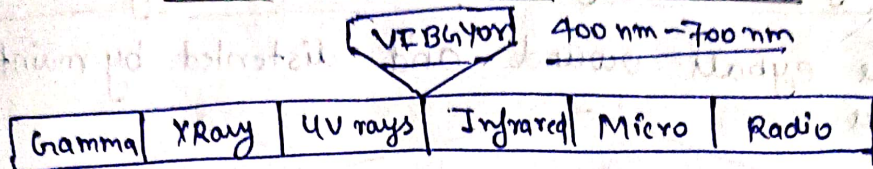
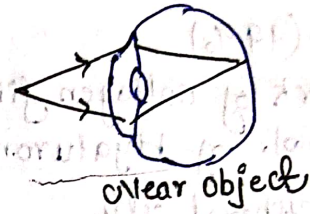
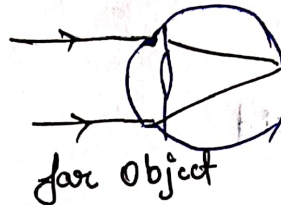
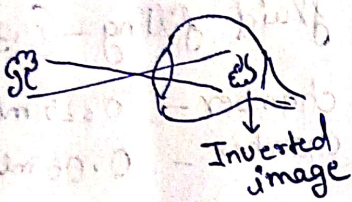
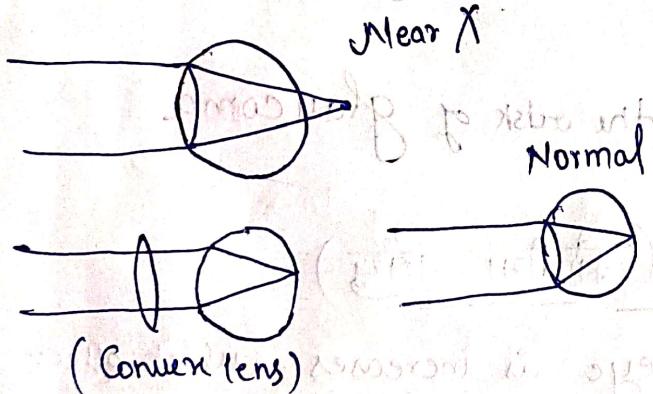


Image formation

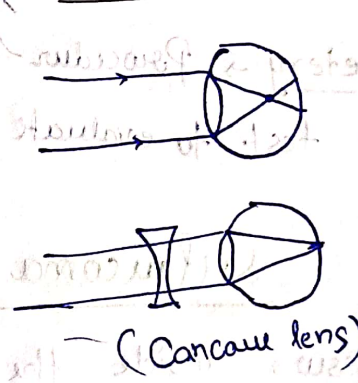


Class - 4

Far Sightedness (Hyperopia)



Near Sightedness (Myopia)

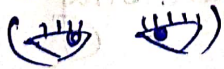


Refractive Power of lens

$$D = \frac{1}{f} = RP$$

Convex lens $\rightarrow +ve$
Concave lens $\rightarrow -ve$

Bimocular vision - vision using 2 eyes with overlapping field of view.



Fluid System of eye - Intraocular fluid

↓
Keep the eyeball round and distended by maintaining pressure

Vitreous

Aqueous Humour (RI $\rightarrow 1.34$)

(In front of lens)

Water (99%)

Network of collagen fibrils

Large mol. of Hyaluronic acid

Peripheral cells

Inorganic salts, Sugar, Ascorbic acid

Aqueous

(1.34)

Vitreous Humour

(Back of lens)

Watery fluid filling - (0.31)

Anterior chamber - 0.25 ml

Posterior chamber - 0.06 ml

Density - 1.025 - 1.040

pH = 7.2 (acidic)

ROF = 2.3 μ l/min.

Tonometer $\rightarrow$ Measure the fluid pressure (in eye)

Ocular Tonometry $\rightarrow$ Procedure

Important test to evaluate the risk of glaucoma

Glaucoma (नीतीमा बिद्यु)

The fluid pressure inside the eye is increases which affect the optic nerve. Due to which, the vision decreases.
(axons of optic nerve compressed)

Blockage of Cytoplasm.

Lack of appropriate nutrition.

Treatment - : ① Laser Trabeculoplasty
② Trabeculectomy

Cataract :- a cloudy area in the lens of your eye.

X (helps to focus)



Normal



with cataract

Protein and fibers in lens are break down.

(age $\uparrow$)

Photo-Receptors - Rods & Cones

(126 million)

126 million

6 million

Rods - Responsible for Night Vision. (Boundary of retina)

Cones - Provide colour vision in brighter light. (Centre of retina)
(Green, Red, Blue)

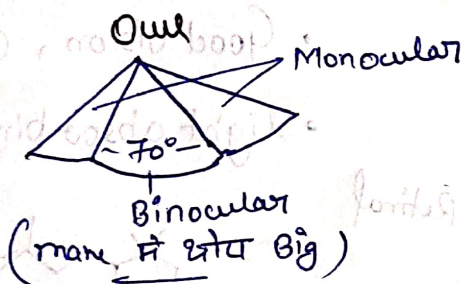
(No colour in dim light because the cone cells, which detect colour, are relatively insensitive to light.)

Rods -  → Rhodopsin pigment (vit A) (lack - Night Blindness)
Cones -  → Iodopsin pigment (Chromophore) (lack - colour Blindness)

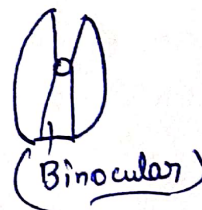
Photoreceptor pass info. through graded pot. Not Action pot.
(Vary cont. amplitude)

Bird Vision

(Good eyesight)



Pigeon



Reptile - Avian eye. (Can change the shape of lens).

Nocturnal Birds → tubular eyes, (low no. of colour detectors, high density of rod cells.)

Class - 5

Visual Cycle (George Wald) (Also called Wald's Visual cycle)

Process by which light $\rightarrow$ electric signals in rod, cones and photosensitive ganglion cell of retina.

Also called - Visual phototransduction.

Photopigments

Unstable pigment that undergo a chemical change when they absorb light.

Opsin

(Glycoprotein)

(4 types)

3 in Cone 1 in Rods

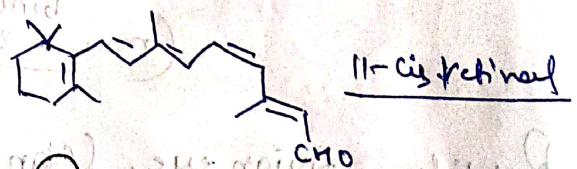
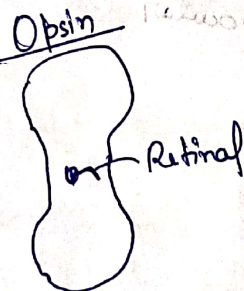
Retinal

Found in both

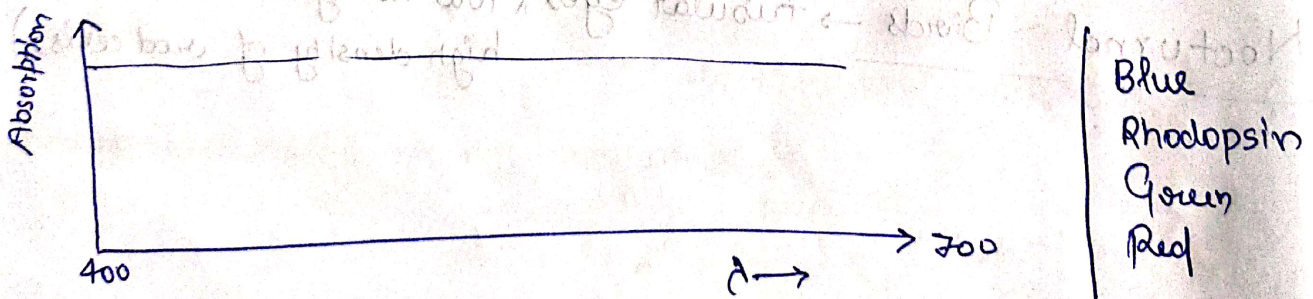
(Derivative of Vitamin A)

formed by Carotene (orange color)

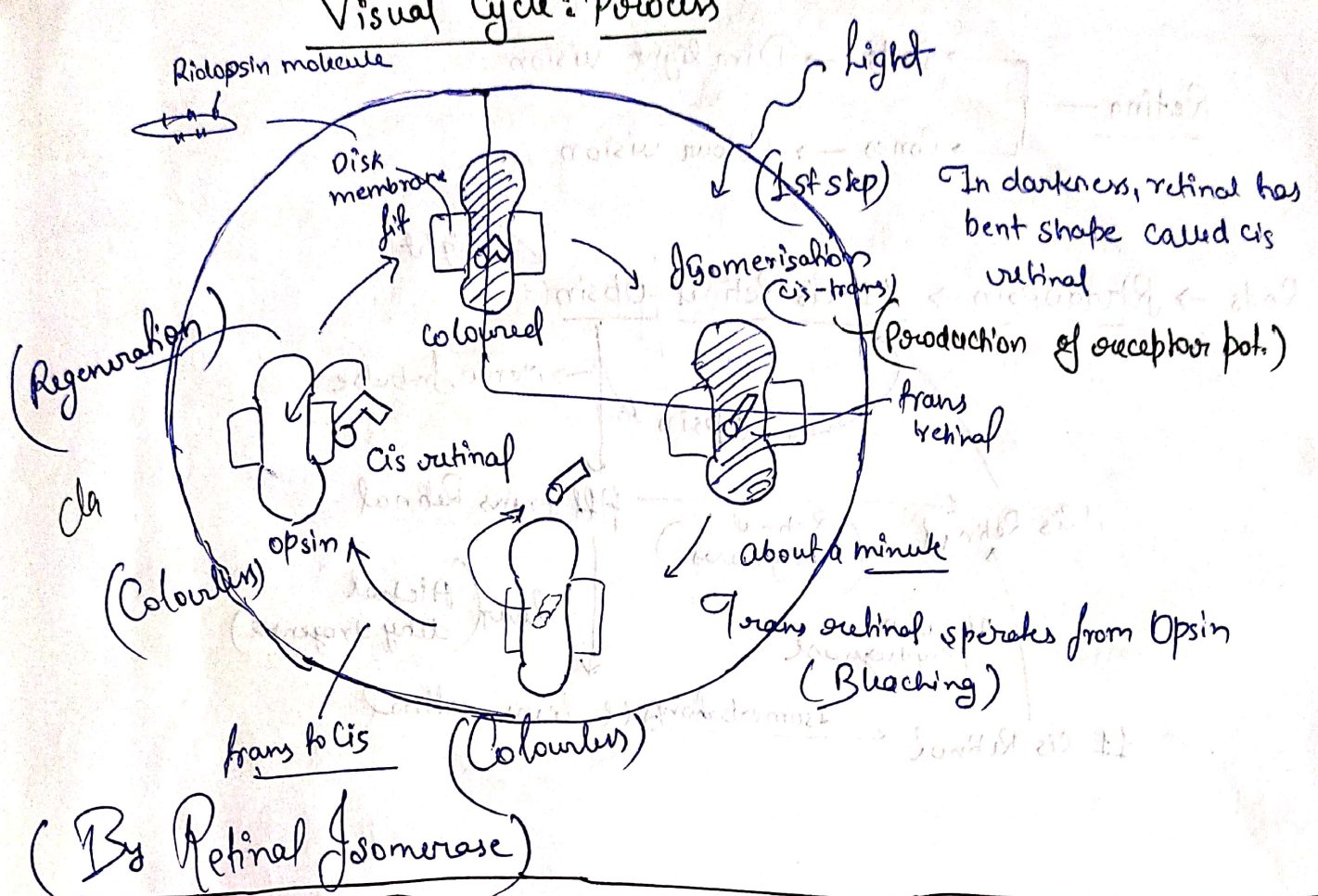
- Good vision, Carotene $\uparrow$,
- light absorbing part



Absorption spectra of Human Visual Pigments -



Visual Cycle: Rhodopsin



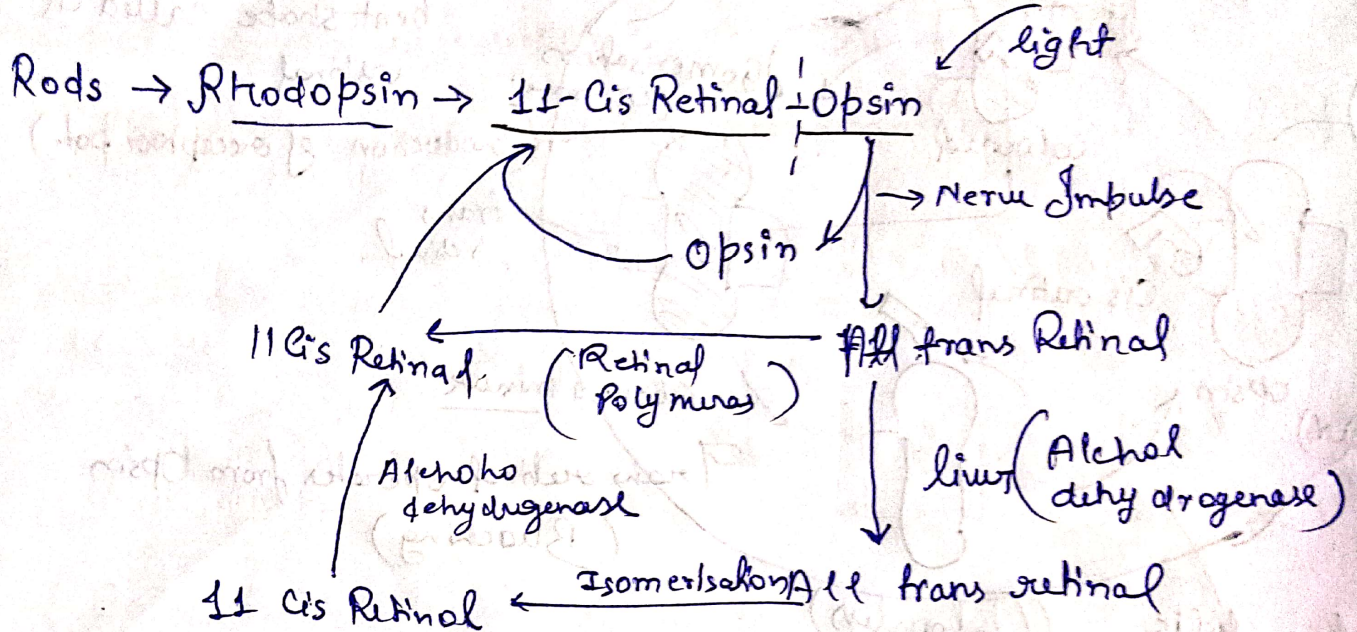
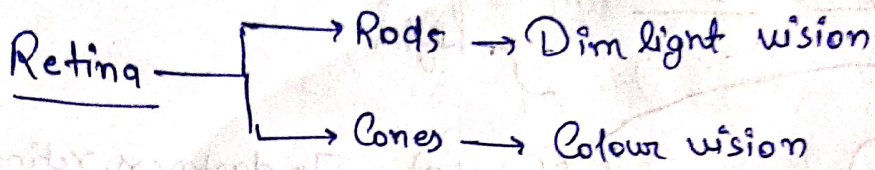
Dark & Light adaption

Light adaption: Dark $\longrightarrow$ Light | Dark adaption: Light $\longrightarrow$ Dark
 (Fast) (in few sec.) | (~~Dark~~) Slow (in min.)

(Important)

- ① The pigmented layer of the retina adjacent to the photoreceptors stores a large quantity of Vit A and contributes to the regeneration in rods.
- ② The extent of rhodopsin generation decreases drastically if the retina detaches from the pigmented layer.
- ③ Cone photopigments regenerate much more quickly than rhodopsin in rods (less dependent on the pigmented layer).

Half of all rod pigments take 5 min. to regenerate.
 Cone pigments take 90 sec. only.

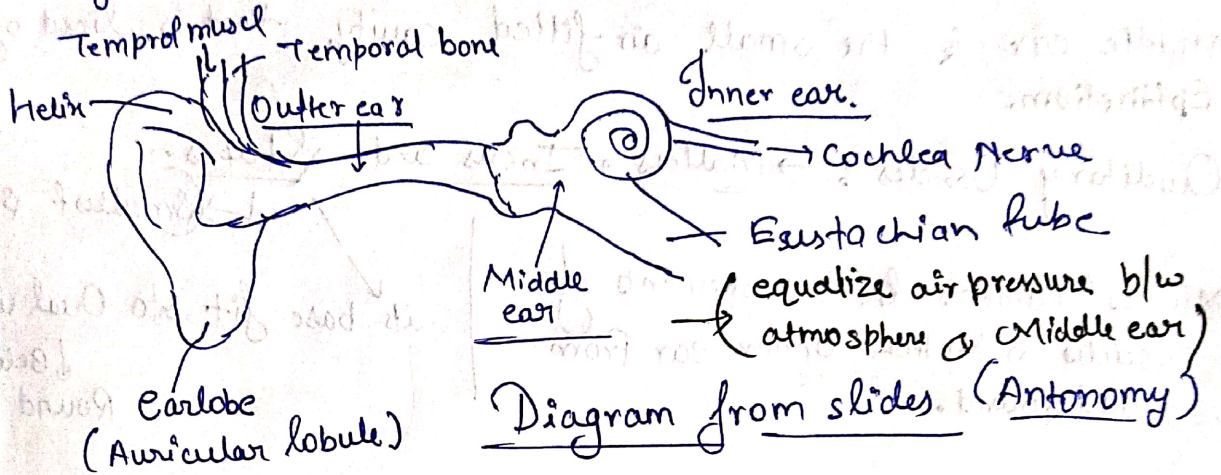


Ear → Convert Mechanical energy → Electrical energy. signals

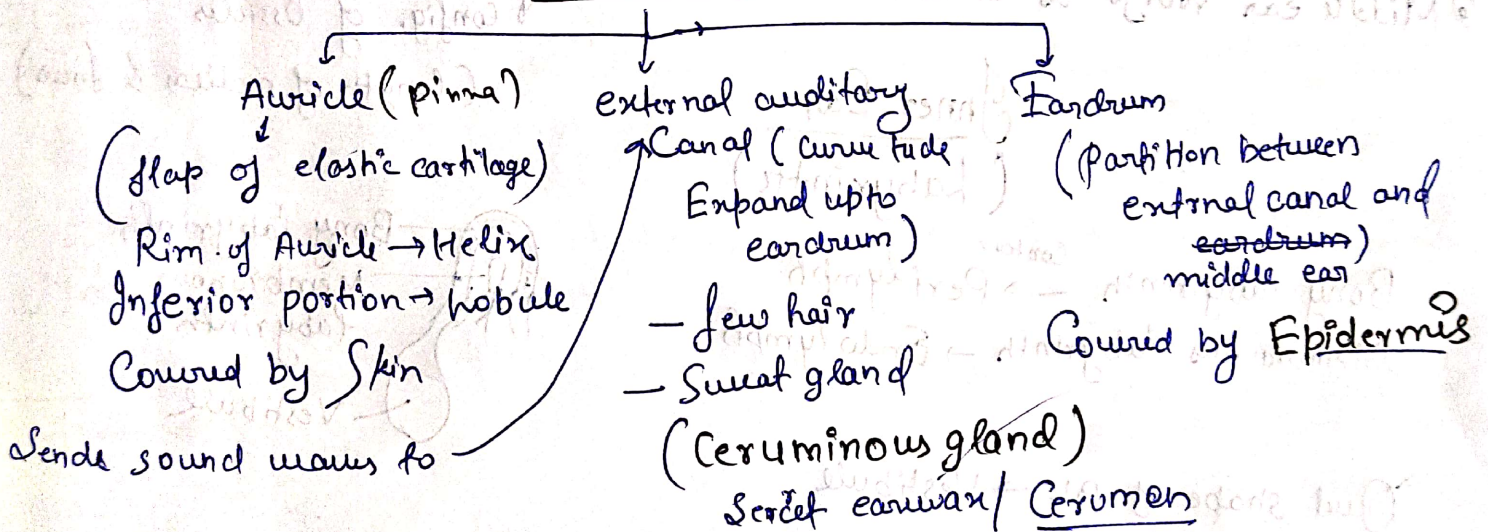
↓
Contains receptor for equilibrium.

↓
1000x faster than photoreceptor
Can respond to light.

- gold atom (0.3 nm) into BS.



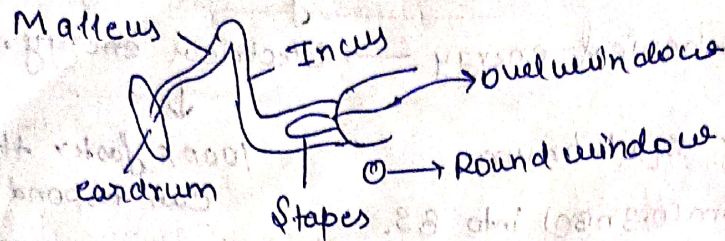
Outer Ear



Perforated eardrum — A hole in the tissue that separates the outer ear from the middle ear. Caused by loud sound, head trauma, a foreign object in the ear.

Impacted Ceruman — Earwax Blockage, Ear wax clear your ear.
Block your ear canal.

Middle ear



(Stapes vibrates oval window ~22,100 Hz, more than eardrum)

Middle ear is the small air-filled cavity that is lined by the Epithelium.

★ Auditory Ossicles: → Malleus, Incus and Stapes.

↳ (Smallest skeletal muscle)

Muscle/Ligaments limits movements of ossicles to protect inner ear from loud sound.

its base fit into Oval window
↓ Below Round window

① Middle ear works as an amplifier in 2 ways

→ Area diff (55 to 3.2 mm²)

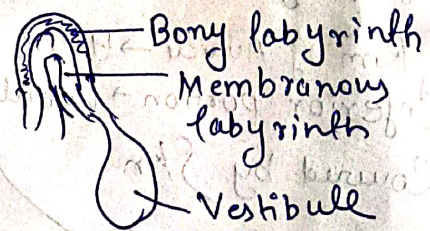
→ Config. of Ossicles

(length of malleus & Incus)

Inner Ear (Labyrinth)

Bony labyrinth → Perilymph

Membranous labyrinth → Endolymph



Oval shape of BL → Vestibule

Utricle (little bag)

Saccul (little Sac)

Connected to

Ampulla

• Vestibule houses sensory as well as motor neurons that synapse with receptors for "equilibrium".

• Anterior to Vestibule is Cochlea, a bony spiral canal.

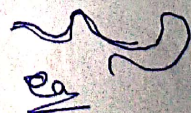
(participate in process of auditory transduction)

• The cochlea conducts sound through liquid not air.

This liquid has a much higher inertia than air i.e. it is harder to move.

Hence, more pressure is required.

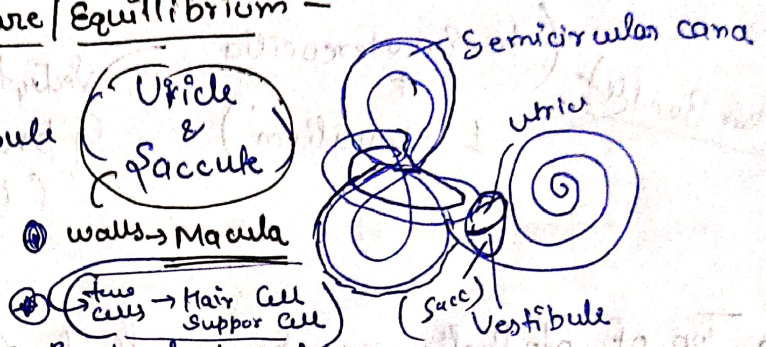
Read Physiology from Slides.



Equilibrium (Balance) $\longrightarrow$ Vestibular System $\longrightarrow$ Inner Ear
 Sound $\longrightarrow$ Cochlea

Physiology of Balance / Posture / Equilibrium -

Semicircular Canal and Vestibule concerned with Balance



3 Semicircular Canals $\longrightarrow$ at 90° that perceive position of head in space, (roll, pitch, yaw axis)

Change in position of head cause movement of fluid $\begin{cases} \text{Perilymph (Outside)} \\ \text{Endolymph (inside)} \end{cases}$

(Hair Bundle)
 40-80 Stereocilia + 1 Kinocilia

Hair cells (Sensory receptors) present in Utricle, Saccule and Ampulla
 bends and stimulates. (Small elevation Crista)

Nerve impulses generated and join the vestibule part of Vestibulochochlear nerve.

Transmits to Cerebellum, then Cerebrum.

Crista covered by gelatinous material (Cupula)

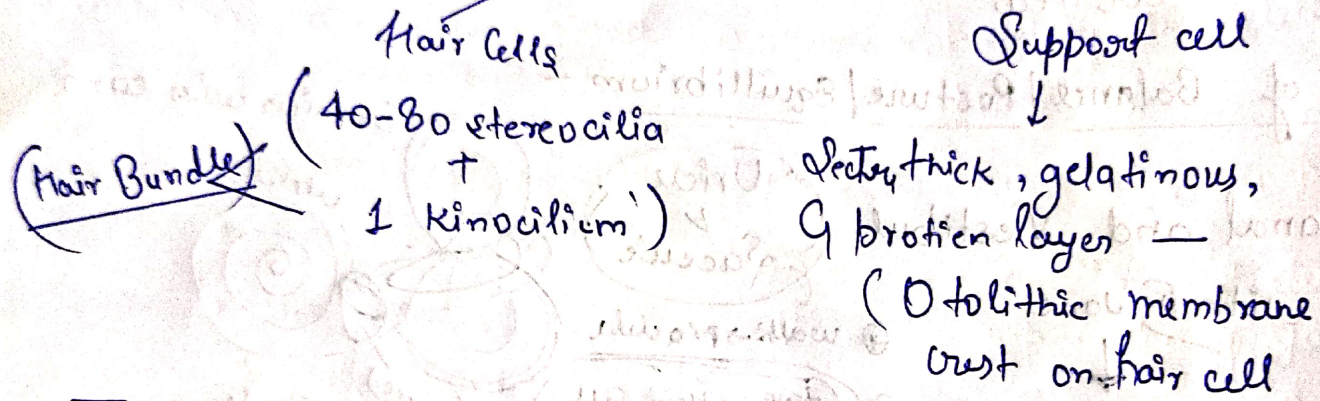
Maintain Balance / Posture.

• Lower frequency $\longrightarrow$ Entire Basilar membrane does not vibrate simultaneously

2 types of Balance

- Static Balance - (mainly head), Tilting of head and linear acc./decc.
 (receptors)
 2 maculae ($\downarrow$)
- Dynamic Balance - Refers to the maintenance of pos - (mainly head) is response to sudden movement, (rot. acc. or decc.)
 (3 ducts)
 Semicircular

• Wall of Utricle & Sacula → Macula



• Top of otolithic membrane floats CaCO_3 crystals called Otoliths.

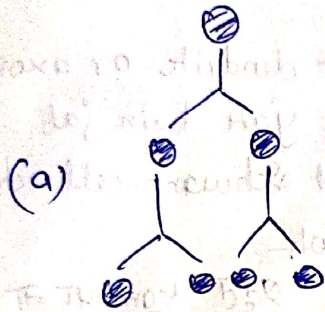
neurons are found in group.

Neural Circuits

Circuit of Billions of neurons in CNS.

Simpler neural circuit → One Series Combination.
process specific type of info.

Neuron

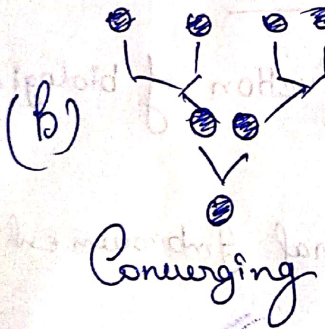


A Single presynaptic neuron may synapse with several post synaptic neurons.

This amplifies the signal.

Diverging Circuit

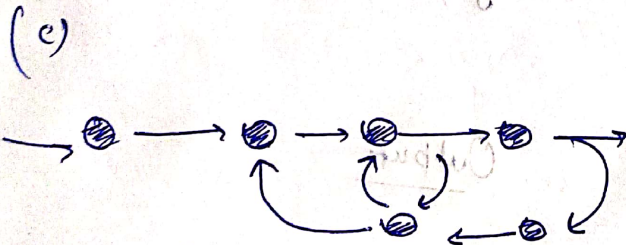
One neuron from brain can activate many motor nerves.



Several presynaptic neurons synapse with single post synaptic neuron.

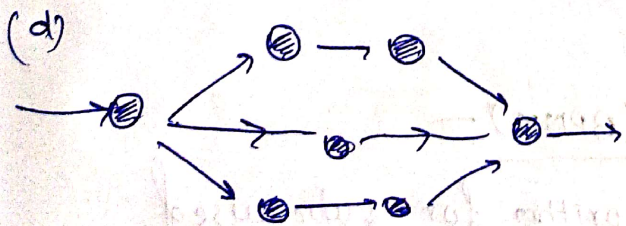
Sensory info. travels to the Brain in this way

Converging



Reverberating

Controls rhythmic activity (breathing)
(walking)
(Waking Up)



A Single presynaptic cell stimulates a group of neurons, each of which synapses with a common postsynaptic cell.

Eg - Coincidence detection.

Math calculation.

(Parallel after discharge)

Plasticity & Regeneration

Capabilities to change based on experience.

Included → Sprouting of new dendrites.

- Synthesis of new proteins.

- Both chemical & electrical

Signals drive these changes

Capability to replicate/repair themselves.

Mammalia neuron → limited capacity

In PNS, the dendrite or axon repair ho y jate hain jab cell body and Schwann cell active ho. Nahi toh

CNS me toh $\frac{1}{2}$ km ya 1 m ki Barabar repair hote hain

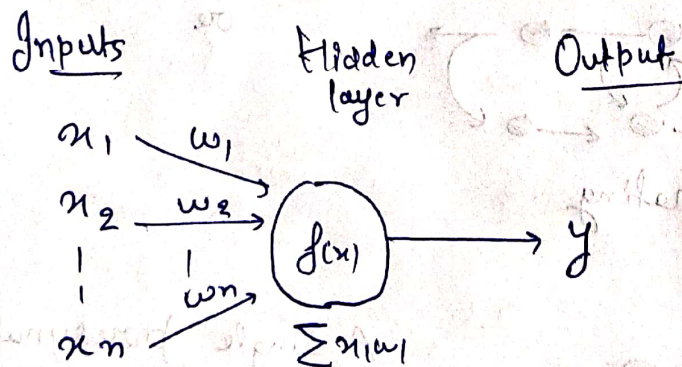
Artificial Neural Network (ANN)

Computational model based on the structure and function of biological neural networks. (Brain = Machines).

Data → Meaning → hearing and neural improvement

(Learning in a sense to on itself)

Usage - ML AI



Back-Propagation Neural Network (BPNN) -

(Backward propagation of error) → algorithm for supervised learning of artificial neural networks using gradient descent.

- Calculates the gradient of error function wrt. neural network weight (w_1, w_2 →)
- (b → bias value)